

An Aerial and Ground Base Station Cooperation Strategy for UAV and Cellular Integrated Networks

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Abstract—This article proposes an aerial and ground base station cooperation (AG_CoMP) strategy for improving the downlink transmission performance of an aerial UAV user (AUE) in an unmanned aerial vehicle and cellular integrated network. The AG_CoMP strategy allows both a ground base station (GBS) and an aerial base station (ABS) to participate in BS cooperation, and seeks to reduce the negative impact of BS cooperation on the overall network performance. The GBS and ABS closest to an AUE are associated to cooperatively provide downlink transmission of the AUE and a blanking area is defined to disallow nearby GBSs to transmit during the transmission of the AUE. A combination of noncoherent joint transmission and dynamic point blanking is used to reduce the large signaling overhead for transmission of the AUE's data between the cooperative BSs with the proposed strategy. Performance models are derived to analyze the downlink transmission performance of an AUE in terms of the coverage probability, achievable throughput, and SINR meta distribution of the AUE. Numerical results show that compared with three benchmark strategies, the proposed AG_CoMP strategy can ensure a relatively good downlink transmission performance, and achieve a better downlink transmission performance when an AUE is at an intermediate altitude between an ABS and a GBS.

Index Terms—Base station, cooperation strategy, UAV and cellular integrated network (Ucell_Net), unmanned aerial vehicle (UAV).

I. INTRODUCTION

WITH the development of unmanned aerial vehicle (UAV) technologies, the integration of UAVs and traditional cellular networks has enabled a variety of civilian and military applications, which has aroused a great attention to UAV and cellular integrated networks (Ucell_Net) from both industry and academia [1]. In a Ucell_Net, a UAV can be used as an aerial base station (ABS) in some hotspot

areas (e.g., large gathering venues or large athletic events) to provide uplink and downlink data transmission for ground cellular users (GUEs) or aerial UAV users (AUEs) to alleviate the transmission load of ground base stations (GBSs) [2]. It can also be used to execute an aerial task as an AUE, such as geological prospecting, forest fire prevention, and forestry plant protection [1]. Compared with a traditional cellular network, a Ucell_Net is an interference-limited network where an AUE may suffer more severe interferences from nearby GBSs or ABSs [2], such as the intercell interference, caused by a GBS and the Line-of-Sight (LoS) interference caused by an ABS. Such severe interferences have become a major issue restricting the transmission performance of the network.

To tackle this issue, BS cooperation can be introduced to mitigate the interferences and provide reliable link transmission for AUEs [3], [4]. In this context, considerable studies have been conducted and a variety of BS cooperation strategies have been proposed for cellular networks. However, most of those studies only considered GBSs in BS cooperation, which may not be efficient for a Ucell_Net. In such networks, an AUE typically operates at a high altitude. The distance between an AUE and a GBS has a big effect on the cooperation gain of GBSs and accordingly the transmission performance of the AUE. Compared with a GBS, if an ABS closer to an AUE participates in BS cooperation, it may bring a larger cooperation gain for the AUE. Thus, it is interesting to study an aerial and ground BS cooperation strategy in which GBSs and ABSs can cooperatively provide a link transmission for an AUE at a high altitude to improve the transmission performance for the AUE.

In this article, we study BS cooperation in a Ucell_Net and propose an aerial and ground BS cooperation (AG_CoMP) strategy for improving the downlink transmission performance of an AUE in the network. The proposed AG_CoMP strategy allows both a GBS and an ABS to participate in BS cooperation, and seeks to reduce the negative impact on the overall network performance. The GBS and ABS closest to an AUE are associated to cooperatively provide downlink transmission of the AUE and a blanking area is defined to disallow nearby GBSs to transmit during the transmission of the AUE. A combination of noncoherent joint transmission and dynamic point blanking (DPB) is used to reduce the large signaling overhead for transmission of the AUE's data between the cooperative BSs with the proposed strategy. Performance models are derived to analyze the downlink transmission performance of an AUE in terms of the coverage probability, the achievable throughput, and the signal to interference plus

Manuscript received 15 July 2023; revised 30 September 2023; accepted 16 October 2023. Date of publication 20 October 2023; date of current version 7 March 2024. This work was supported in part by the Open Research Projects of Zhejiang Lab under Grant 2021LC0AB02, and in part by the Fundamental Research Funds for the Central Universities under Grant 2242022k60001. (Corresponding author: Jun Zheng.)

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Digital Object Identifier 10.1109/IJOT.2023.3326316

noise ratio (SINR) meta distribution of the AUE. Simulation results are shown to justify the effectiveness of the derived performance models, and numerical results are presented to evaluate the performance of the proposed AG_CoMP strategy.

The remainder of this article is organized into six sections, including related work, a network model, an AG_CoMP strategy, performance analyses, performance evaluation, and conclusions.

II. RELATED WORK

BS cooperation has been extensively studied for cellular networks and a variety of BS cooperation strategies have been proposed to improve the transmission performance of UEs for a variety of cellular networks [5], [6], [7]. However, related work on BS cooperation for a UAV cellular network is still limited. Sun et al. proposed a user-centric cooperative scheme to serve the GUEs in a malfunction area [8]. According to the proposed scheme, the malfunction area is partitioned into a UAV area, a cooperation area and the nearest GBS area. Mirbolouk et al. considered a hybrid satellite-UAV relay network, where UAVs as aerial BSs use a joint transmission coordinated multipoint (CoMP) strategy to provide downlink transmission for nonorthogonal multiple access (NOMA) GUEs, and analyzed the outage probability of the network [9]. Hajiakhondi-Meybodi et al. used a CoMP transmission strategy to mitigate the intercell interference received by a GUE in a UAV-aided cellular network, taking into account indoor and outdoor caching service scenarios [10]. Tang et al. proposed a 3-D coordination model for downlink transmission of a GUE via multicell beamforming and signal cooperation in a multiantenna UAV network [11]. The above work only considered BS cooperation for a GUE in a UAV cellular network, not for an AUE. Compared to a GUE, an AUE usually needs to fly or hover at different altitudes for executing different aerial tasks, which should be taken into account in the design of a BS cooperation strategy.

In the context of BS cooperation for an AUE, Li and Xia designed a CoMP transmission scheme based on Poisson–Delaunay triangulation to provide reliable and seamless connectivity for UAVs, and proposed a dynamic frequency allocation scheme to eliminate intercell interference [12]. Amer et al. introduced a CoMP transmission strategy to improve the transmission performance of an AUE in a cellular-connected UAV network, and derived the coverage probability of the AUE for static and mobile scenarios [13]. In our previous work [3], [4], we used a fixed-number GBS cooperation scheme based coherent and noncoherent joint transmission to improve the coverage probability, achievable throughput and rate meta distribution of an AUE. Sun et al. studied the strongest average received signal strength (RSS)-based CoMP scheme for uplink transmission in a cellular-connected UAV network, and developed an analytical framework to evaluate the coverage probability and area spectral efficiency of the network [14]. Li et al. proposed a 3-D UAV-BS cooperation strategy to enhance the transmission quality of an AUE and derived the coverage probability and achievable data rate of the AUE in the general and worst cases [15]. However, all the

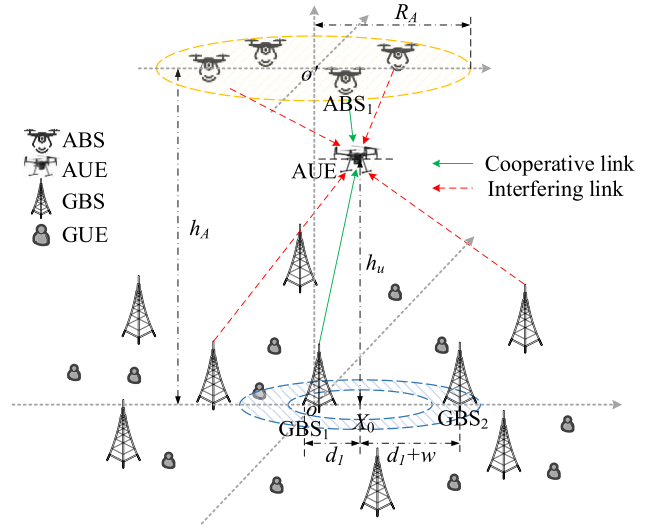


Fig. 1. Network scenario.

above work only considered BS cooperation between GBSs or ABSs for an AUE in a cellular-connected UAV network or a 3-D UAV network, but did not consider the cooperation between a GBS and an ABS, which may not be efficient for a Ucell_Net.

III. NETWORK MODEL

This section describes the network scenario, and introduces the antenna gain models and channel models considered in the network scenario.

A. Network Scenario

Consider a Ucell_Net composed of GBSs, GUEs, and ABSs and AUEs, as shown in Fig. 1, where the GBSs and ABSs provide downlink and uplink transmission for the AUEs or GUEs. In particular, an AUE can be provided reliable link transmission jointly by a GBS and an ABS for its task data. For notational convenience, we establish a 3-D Cartesian coordinate system. The aerial origin of the coordinate system is set at o' , which is the center of a circular ABS distribution area (i.e., the orange shaded area). The ground origin of the coordinate system is set at o , which is the projection of the aerial origin.

Assume that there are N ABSs uniformly distributed in the ABS distribution area with radius R_A . According to [2] and [16], the transmission performance of an aerial network where ABSs are uniformly distributed in a given altitude range perfectly matches that of an aerial network where ABSs are uniformly distributed at an average altitude of the corresponding altitude range. Thus, we assume that all ABSs are located at the same altitude h_A . Let $\Phi_A = \{x_1, x_2, \dots, x_i, \dots, x_N\}$, $i = 1, 2, \dots, N$ denote the 2-D spatial distribution of the ABSs, where x_i denotes the spatial location of the i th ABS. According to stochastic geometry [17], we assume that Φ_A follows a 2-D Binomial point process (BPP) because a BPP distribution can effectively model the distribution of a given number of ABSs whose locations exhibit randomness or uncertainty [2], [16].

Also, the GBSs are uniformly distributed on the ground at an altitude of h_G . Let $\Phi_G = \{y_1, y_2, \dots, y_j, \dots\}, j = 1, 2, \dots$, denote the 2-D spatial distribution of the GBSs, where y_j denotes the spatial location of the j th GBS. According to [3], we assume that Φ_G follows a 2-D homogeneous Poisson point process (HPPP) with density λ_G . Moreover, let h_u denote the altitude of an AUE and assume that $h_G < h_u < h_A$.

Also assume that each ABS is connected to a GBS through a perfect wireless backhaul link [2], [16]. Each GBS or ABS shares the same orthogonal time-frequency resource blocks to provide downlink transmission for both GUEs and AUEs within its coverage, which is called orthogonal spectrum sharing (OSS) in [2]. Thus, for an AUE or GUE, only the intercell interferences from GBSs and ABSs need to be considered. Moreover, we assume that all GBSs and ABSs are synchronized, which can be implemented by using available synchronization techniques [5], [7].

According to [16] and [17], the distribution of the ABSs in area $b(o', R_A)$ is isotropic and rotation invariant around the aerial origin o' , where $b(o, x)$ denotes a circular 2-D area with o as the center and x as the radius of the area. Thus, we will focus on the downlink transmission performance of a typical AUE located at $\mathbf{x} = (X_0, 0, h_u)$ over the X-axis (i.e., “AUE” in Fig. 1).

B. Antenna Gain Model

Assume that each AUE uses an omnidirectional antenna, and each GBS uses vertically directional and horizontally omnidirectional antenna [2], [13], [18], [19]. The downlink transmissions of an AUE and a GUE are provided by the sidelobes and the mainlobe of a GBS’ antenna, respectively [13]. Let g_G denote the sidelobe gain of a GBS’ antenna. For an ABS, we assume that it uses a directional antenna and we use the conic directional antenna model [20] to characterize the antenna gain of an ABS, i.e.,

$$g_A = g_A^{\max} \cos(\varphi)^\omega \quad (1)$$

where $g_A^{\max}$, $\varphi \in [0, (\pi/2)]$, and ω denote the maximum gain, the incidence angle, and the directivity factor of an ABS’ antenna. A larger value of ω represents a narrower conic antenna. As shown in Fig. 2, the incidence angle of an ABS’ antenna is given by

$$\varphi = \arccos\left(\frac{\Delta h_{AU}}{\sqrt{\Delta h_{AU}^2 + d_i^2}}\right) \quad (2)$$

where $\Delta h_{AU} = h_A - h_u$ denotes the altitude difference between an ABS and an AUE, d_i denotes the horizontal distance between the i th ABS and an AUE.

C. Channel Model

For both the ground-to-air (G2A) channel between a GBS and an AUE, and the air-to-air (A2A) channel between an ABS and an AUE, we assume that they are time-invariant during a single time-frequency resource block [2], [13], [18]. Thus, we only need to consider the large-scale fading, small-scale fading, and the LoS probability of a G2A channel or an A2A channel.

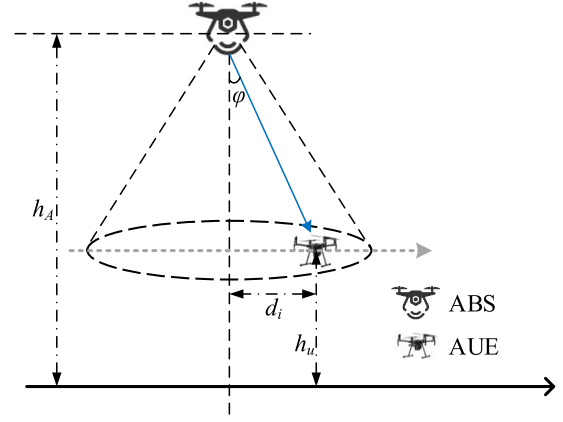


Fig. 2. Illustration of the antenna incidence angle of an ABS.

According to [3], the large-scale fading of the channel between a GBS and an AUE is characterized by the following model:

$$\xi_v(d_j) = \rho_v \left[d_j^2 + \Delta h_{UG}^2 \right]^{-\frac{\alpha_v}{2}}, v \in \{L, \bar{L}\} \quad (3)$$

where d_j denotes the horizontal distance between the j th GBS and the AUE, $\Delta h_{UG} = h_u - h_G$ denotes the altitude difference between a GBS and the AUE, ρ_v denotes the path loss of the channel at a reference distance of 1, α_v is a large-fading parameter of the channel, $v \in \{L, \bar{L}\}$ indicates the state of the channel, i.e., $v = L$ and $v = \bar{L}$ indicate that the channel between a GBS and an AUE is LoS and Non-LoS (NLoS), respectively.

For the small-scale fading of the channel, it is characterized by a Nakgami-m fading model. According to [3], the power gain of the small-scale fading (Ω_v) follows a gamma distribution with parameter m_v , i.e., $\Omega_v \sim \text{Gamma}(m_v, [1/m_v])$. The cumulative distribution function (CDF) of the gamma distribution is given by

$$F_{\Omega_v}(x) = P[\Omega_v < x] = 1 - \sum_{k=0}^{m_v-1} \frac{(m_v x)^k}{k!} \exp(-m_v x). \quad (4)$$

Moreover, according to [2], the LoS probability of the channel between a GBS and an AUE can be obtained by using the “Curve Fitting” toolbox in MATLAB, i.e.,

$$P_{\text{LoS}}(d_j) = -a_{\text{LoS}} \exp\left(-b_{\text{LoS}} \tan^{-1}\left(\frac{\Delta h_{UG}}{d_j}\right)\right) + c_{\text{LoS}} \quad (5)$$

where $(a_{\text{LoS}}, b_{\text{LoS}}, c_{\text{LoS}})$ is a set of parameters via curve fitting, and different values of $(a_{\text{LoS}}, b_{\text{LoS}}, c_{\text{LoS}})$ correspond to different transmission environments. Based on (5), the NLoS probability of the channel between the j th GBS and an AUE is given by $P_{\text{NLoS}}(d_j) = 1 - P_{\text{LoS}}(d_j)$. Thus, Φ_G can be divided into two subsets Φ_G^L and $\Phi_G^{\bar{L}}$, i.e., $\Phi_G = \Phi_G^L \cup \Phi_G^{\bar{L}}$, where Φ_G^L represents a set of GBSs that have an LoS downlink to a typical AUE, and $\Phi_G^{\bar{L}}$ represents a set of GBSs that have an NLoS link to the AUE. According to [3], Φ_G^L is a PPP with density λ_G^L , and $\Phi_G^{\bar{L}}$ is a PPP with density $\lambda_G^{\bar{L}}$, where $\lambda_G^L = P_{\text{LoS}} \cdot \lambda_G$ and $\lambda_G^{\bar{L}} = P_{\text{NLoS}} \cdot \lambda_G$.

For the A2A channel, we assume no obstacle between the ABS and the AUE, i.e., the channel is a LoS channel. Thus, the large-scale fading of the channel is given by

$$\xi_A(d_i) = \rho_L \left[d_i^2 + \Delta h_{AU}^2 \right]^{-\frac{\alpha_L}{2}}. \quad (6)$$

Similar to the G2A channel between a GBS and an AUE, the small-scale fading of the A2A channel is also characterized by a Nakagami-m fading model. According to [2], the power gain of the small-scale fading (Ω_A) also follows a gamma distribution with parameter m_L .

IV. AERIAL AND GROUND BS COOPERATION STRATEGY

This section presents the proposed AG_CoMP strategy for a Ucell_Net.

In a Ucell_Net, an AUE typically executes a task in the air. The distance between an AUE and a GBS has a big effect on the transmission performance of the downlink between the GBS and the AUE. If only GBSs participate in BS cooperation, the cooperation gain would be largely reduced if the AUE is located at a high altitude. Similarly, an ABS also operates in the air. Compared with a GBS, if an ABS closer to an AUE participates in BS cooperation, it may bring a larger cooperation gain for the AUE. Thus, a BS cooperation strategy should allow an ABS to participate in BS cooperation. According to [3] and [5], the increase in the cooperative gain of a BS cooperation strategy gradually decreases with the increase in the number of cooperative BSs. Thus, the number of BSs participating in BS cooperation should not be too large. On the other hand, a BS cooperation strategy with a large number of cooperative BSs may have a negative impact on the overall network performance. For these reasons, the proposed AG_CoMP strategy should limit the number of cooperative ABSs and GBSs.

According to [6], BS cooperation based on geographical location would produce less signaling messages than that based on signal power. Thus, it is preferred to use a closest distance association strategy in BS cooperation. Moreover, according to 3GPP technical report [21], BS cooperation using coherent joint transmission needs to exchange more channel state information (CSI) compared with that using noncoherent joint transmission [5]. According to [5] and [6], noncoherent joint transmission is more robust to channel uncertainties and synchronization errors than coherent joint transmission. Thus, it is preferred to use noncoherent joint transmission. On this basis, to reduce the severe overhead of exchanging signaling messages and AUE data (e.g., the X2 interface described in [22]) in the network, it is preferred to use a combination of noncoherent joint transmission and DPB, which does not require user data sharing between adjacent BSs participating in DPB and thus can avoid strong interference from adjacent BSs by muting certain time-frequency resource blocks of adjacent BSs [23].

Based on the above considerations, we propose an AG_CoMP strategy for a Ucell_Net, which has the following characteristics.

- 1) An ABS is allowed to participate in the BS cooperation for a typical AUE, and the number of ABSs participating in the BS cooperation should be limited.
- 2) A closest distance association strategy is used to associate a GBS and an ABS with an AUE in BS cooperation for downlink transmission of the AUE.
- 3) A combination of noncoherent joint transmission and DPB is used to reduce the large signaling overhead between cooperative BSs and mitigate severe GBS interferences.

Specifically, for a typical AUE, its downlink transmission is performed through BS cooperation, and the GBS and ABS closest to the AUE are associated as cooperative BSs to participate in the BS cooperation. Obviously, more ABSs and GBSs can be used to cooperatively provide downlink transmission for the AUE. But to reduce the negative impact of BS cooperation on the overall network performance, we only consider the cooperation between the closest GBS and the closest ABS to the AUE in the proposed AG_CoMP strategy. The optimal number of cooperative ABSs and GBSs depends on several network conditions, such as the maximum number of AUEs allowed to access the network, the interference signal power received by an AUE, and the overall load of the network. The optimization of the number of cooperative ABSs and GBSs is not the focus of this work and will be studied in our future work.

To mitigate the interferences caused by the GBSs close to the AUE, we define a blanking area on the ground, in which all GBSs are not allowed to use the time-frequency resource blocks being used by the cooperative GBS through the management of their time-frequency resources [24]. The blanking area is a ring area centered on the projection location of the AUE on the ground. The radius of the inner ring area equals to the horizontal distance between the closest GBS and the AUE, and the radius of the outer ring area equals to the inner ring radius plus a given ring width.

The two cooperative BSs constitute a set, which is denoted by

$$\Phi_C = \{x_1, y_1 | x_1 \in \Phi_A, y_1 \in \Phi_G\} \quad (7)$$

where x_1 denotes the 2-D location of the closest ABS to the AUE and y_1 denotes the 2-D location of the closest GBS to the AUE. The GBSs located in the blanking area of the AUE is called blanking GBSs of the AUE, which are denoted by

$$\Phi_B = \{y_j | y_j \in b(\mathbf{x}', \|y_1\| + w) \setminus b(\mathbf{x}', \|y_1\|), y_j \in \Phi_G\} \quad (8)$$

where $\mathbf{x}'$ denotes the ground projection of $\mathbf{x}$, i.e., $\mathbf{x}' = (X_0, 0, 0)$, $\|y_1\|$ denotes the horizontal distance between the AUE and the closest GBS, and w is the width of the blanking area. As shown in Fig. 1, AUE represents a typical AUE, "GBS₁" and "ABS₁" represent the cooperative BSs of the AUE, "GBS₂" represents a blanking GBS of the AUE, the blue-shaded area represents the blanking area of the AUE.

V. PERFORMANCE ANALYSES

This section derives performance models for analyzing the coverage probability, achievable throughput, and SINR meta

distribution of a typical AUE with the proposed AG_CoMP strategy.

A. SINR

According to [4], we assume that the downlink between a typical AUE and the closest GBS is LoS, and the cooperative ABS can align the antenna's mainlobes with the typical AUE by using a beam alignment technique [2], [24]. Thus, the SINR received at a typical AUE is given by

$$\text{SINR} = \frac{P_{GG}\Omega_L\xi_L(D_G) + P_{AG_A}^{\max}\Omega_A\xi_A(D_A)}{I_G + I_A + \sigma^2} \quad (9)$$

where P_G and P_A denote the transmitting power of a GBS and that of an ABS, respectively; D_G denotes the horizontal distance between the AUE and the cooperative GBS; D_A denotes the horizontal distance between the AUE and the cooperative ABS; σ^2 denotes the additive white Gaussian noise power of a cooperative downlink; I_G denotes a sum of the interference signal power from all GBSs outside the blanking area, and I_A denotes a sum of the interference signal power from all ABSs except the cooperative ABS. Moreover, I_G is given by

$$I_G = \sum_{\Phi_G \setminus \Phi'_B} \varepsilon P_{GG}\Omega_v\xi_v(D_j) \quad (10)$$

where $\Phi'_B = \{y_j | y_j \in \Phi_G \cap b(\mathbf{x}', \|y_1\| + w)\}$ denotes a set of blanking GBSs and cooperative GBS, ε is a 0-1 random variable, indicating whether an interfering GBS is active, and following a Bernoulli distribution with a parameter p_ε , i.e., $\varepsilon \sim B(p_\varepsilon)$, D_j denotes the horizontal distance between the AUE and the j th interfering GBS. I_A is given by

$$I_A = \sum_{\Phi_A \setminus x_1} P_{AG_A}\Omega_A\xi_A(D_i) \quad (11)$$

where D_i denotes the horizontal distance between the AUE and the i th interfering ABS.

B. Distance Distributions

Next we analyze several distance distributions relevant to subsequent performance analyses.

According to [3], the probability density function (PDF) of the horizontal distance between a typical AUE and its closest GBS is given by

$$f_{D_G}(d_G) = 2\pi\lambda_G d_G \exp(-\pi\lambda_G d_G^2). \quad (12)$$

For a typical AUE located over X_0 , let $\{\tilde{D}_i\}$, $i = 1, 2, \dots, N$ denote a set of ordered horizontal distances between the AUE and the ABSs in the network, where i represents the order of the i th closest ABS to the AUE. Obviously, we have $D_A = \min\{\tilde{D}_i\} = \tilde{D}_1$. Since the spatial distribution of the ground projections of the ABSs follows a 2-D BPP, each element in $\{\tilde{D}_i\}$ is independently and identically distributed (i.i.d.) [17]. According to [16] and [25], the conditional CDF of $\tilde{D}_i$ can be obtained based on the overlapping area of $b(\mathbf{x}, \tilde{D}_i)$ and $b(o, R_A)$ divided by the area of $b(o, R_A)$. Since $\tilde{D}_i$ is a random variable, there are two possible cases, depending on the value of $\tilde{D}_i$: 1) $b(\mathbf{x}, \tilde{D}_i)$ falls entirely in $b(o, R_A)$ and 2) $b(\mathbf{x}, \tilde{D}_i)$

partially overlaps with $b(o, R_A)$. As a result, the conditional CDF of each element in $\{\tilde{D}_i\}$ is given by

$$F_{\tilde{D}_i}(\tilde{d}_i|X_0) = P(\tilde{D}_i \leq \tilde{d}_i|X_0) = \begin{cases} F_{\tilde{D}_{i,1}}(\tilde{d}_i|X_0), & 0 \leq \tilde{d}_i \leq R_A - X_0 \\ F_{\tilde{D}_{i,2}}(\tilde{d}_i|X_0), & R_A - X_0 < \tilde{d}_i \leq R_A + X_0 \end{cases} \quad (13)$$

with

$$F_{\tilde{D}_{i,1}}(\tilde{d}_i|X_0) = \frac{\tilde{d}_i^2}{R_A^2} \quad (14)$$

$$F_{\tilde{D}_{i,2}}(\tilde{d}_i|X_0) = \frac{\tilde{d}_i^2}{\pi R_A^2} \left[\theta^*(\tilde{d}_i) - \frac{1}{2} \sin(2\theta^*(\tilde{d}_i)) \right] + \frac{1}{\pi} \left[\phi^*(\tilde{d}_i) - \frac{1}{2} \sin(2\phi^*(\tilde{d}_i)) \right] \quad (15)$$

where

$$\theta^*(\tilde{d}_i) = \arccos\left(\frac{\tilde{d}_i^2 + X_0^2 - R_A^2}{2X_0\tilde{d}_i}\right) \\ \phi^*(\tilde{d}_i) = \arccos\left(\frac{X_0^2 + R_A^2 - \tilde{d}_i^2}{2X_0R_A}\right).$$

Thus, the PDF of each element in $\{\tilde{D}_i\}$ can be derived by taking the derivative of $F_{\tilde{D}_i}(\tilde{d}_i|X_0)$, i.e.,

$$f_{\tilde{D}_i}(\tilde{d}_i|X_0) = \begin{cases} f_{\tilde{D}_{i,1}}(\tilde{d}_i|X_0), & 0 \leq \tilde{d}_i \leq R_A - X_0 \\ f_{\tilde{D}_{i,2}}(\tilde{d}_i|X_0), & R_A - X_0 < \tilde{d}_i \leq R_A + X_0 \end{cases} \quad (16)$$

where $f_{\tilde{D}_{i,1}}(\tilde{d}_i|X_0) = (2\tilde{d}_i/R_A^2)$, $f_{\tilde{D}_{i,2}}(\tilde{d}_i|X_0) = (2\tilde{d}_i/\pi R_A^2)\theta^*(\tilde{d}_i)$.

According to [16], if the typical AUE is located over the origin (i.e., $X_0 = 0$), the CDF and PDF of the horizontal distance between the AUE and each ABS are, respectively, given by

$$F_{\tilde{D}_i}(\tilde{d}_i|X_0 = 0) = \frac{\tilde{d}_i^2}{R_A^2}, \quad 0 \leq \tilde{d}_i \leq R_A \quad (17)$$

$$f_{\tilde{D}_i}(\tilde{d}_i|X_0 = 0) = \frac{2\tilde{d}_i}{R_A^2}, \quad 0 \leq \tilde{d}_i \leq R_A. \quad (18)$$

Based on (13), the conditional CDF of D_A is given by

$$F_{D_A}(d_A|X_0) = P(D_A \leq d_A|X_0) = 1 - P(\min\{\tilde{D}_i\} > d_A|X_0) \\ = 1 - P(\tilde{D}_1 > d_A, \dots, \tilde{D}_N > d_A|X_0) \\ \stackrel{(a)}{=} 1 - [1 - F_{\tilde{D}_i}(d_A|X_0)]^N \quad (19)$$

where (a) follows from the independent and identical distribution property of $\{\tilde{D}_i\}$. The PDF of D_A can be derived by taking the derivative of (19), and is given by (20), shown at the bottom of the next page, where

$$\theta^*(d_A) = \arccos\left(\frac{d_A^2 + X_0^2 - R_A^2}{2X_0d_A}\right) \quad (21)$$

$$\phi^*(d_A) = \arccos\left(\frac{X_0^2 + R_A^2 - d_A^2}{2X_0R_A}\right). \quad (22)$$

If the typical AUE is located over the origin (i.e., $X_0 = 0$), the PDF of D_A is given by

$$f_{D_A}(d_A|X_0=0) = N \left(\frac{2d_A}{R_A^2} \right) \left(\frac{R_A^2 - d_A^2}{R_A^2} \right)^{N-1}, \quad 0 \leq d_A \leq R_A. \quad (23)$$

According to (12) and (20), the spatial distributions of the ABSs and GBSs in the network are independent of each other. Thus, the joint PDF of the horizontal distance between the typical AUE and its cooperative GBS and that between the typical AUE and its cooperative ABS can be expressed as

$$f_{D_A, D_G}(d_A, d_G|X_0) = f_{D_A}(d_A|X_0) f_{D_G}(d_G). \quad (24)$$

According to [25], a set of unordered horizontal distances between the typical AUE and its interfering ABSs is denoted by $\{D_i\}_{i=1:N-1}$, where D_i denotes the horizontal distance between the typical AUE and its i th interfering ABS, and is i.i.d.. Let $\{\tilde{D}_i\}_{i=2:N}$ denote a subset of $\{\tilde{D}_i\}$ excluding the distance between the AUE and the cooperative ABS, i.e., D_A . Thus, the joint PDF of $\{\tilde{D}_i\}_{i=2:N}$ conditioned on X_0 and D_A is given by

$$\begin{aligned} f_{\tilde{D}_i}(d_2, \dots, d_N|D_A, X_0) &\stackrel{(a)}{=} \frac{N! f_{\tilde{D}_i}(d_A|X_0) \prod_{i=2}^N f_{\tilde{D}_i}(\tilde{d}_i|X_0)}{f_{D_A}(d_A|X_0)} \\ &\stackrel{(b)}{=} (N-1)! \prod_{i=2}^N \frac{f_{\tilde{D}_i}(\tilde{d}_i|X_0)}{1 - F_{\tilde{D}_i}(d_A|X_0)} \end{aligned} \quad (25)$$

where (a) follows from the definition of the conditional probability, (b) follows from (16) and (20). According to [25], the term $(N-1)!$ is used to indicate all possible permutations of the elements in $\{\tilde{D}_i\}_{i=2:N}$. Based on (25), the PDF of D_i is given by

$$f_{D_i}(d_i|D_A, X_0) = \begin{cases} \frac{f_{\tilde{D}_{i,1}}(d_i|X_0)}{1 - F_{\tilde{D}_{i,1}}(d_A|X_0)}, & 0 \leq d_A \leq R_A - X_0, d_A \leq d_i \leq R_A - X_0 \\ \frac{f_{\tilde{D}_{i,2}}(d_i|X_0)}{1 - F_{\tilde{D}_{i,1}}(d_A|X_0)}, & 0 \leq d_A \leq R_A - X_0, R_A - X_0 \leq d_i \leq R_A + X_0 \\ \frac{f_{\tilde{D}_{i,2}}(d_i|X_0)}{1 - F_{\tilde{D}_{i,2}}(d_A|X_0)}, & R_A - X_0 \leq d_A \leq R_A + X_0, d_A \leq d_i \leq R_A + X_0. \end{cases} \quad (26)$$

If the typical AUE is located over the origin, (26) can be simplified as

$$f_{D_i}(d_i|D_A, X_0=0) = \frac{2d_i}{R_A^2 - d_A^2}, \quad d_A \leq d_i \leq R_A. \quad (27)$$

C. Conditional Coverage Probability

The conditional coverage probability of a typical AUE is defined as the coverage probability of the AUE conditioned on the spatial locations of the ABSs and GBSs in the network, which can be expressed as

$$P_{\text{cov}|X_0, \Phi_A, \Phi_G}(\delta) = P(\text{SINR} > \delta|X_0, \Phi_A, \Phi_G) \quad (28)$$

where δ denotes an SINR threshold at the typical AUE. The conditional coverage probability represents the probability that the SINR received at an AUE is larger than a given threshold, given the spatial locations of the GBSs and ABSs in the network. According to [7], the conditional coverage probability of the typical AUE is given by (29), shown at the bottom of the next page, where

$$v_k = (-1)^{k+1} \binom{m_L}{k}, \quad \tau = m_L(m_L!)^{-\frac{1}{m_L}} \quad (30)$$

$$s_k = \frac{k\delta\tau}{P_{GG}\xi_L(D_G) + P_{AG}^{\max}\xi_L(D_A)}. \quad (31)$$

Also in (29), (a) follows from that the complementary CDF (CCDF) of a gamma random variable can be approximated as a weighted sum of the CCDFs of exponential random variables [7]; (b) follows from that the expectation about the small-scale fading of active interfering GBSs and interfering ABSs can be expressed as the corresponding Laplace transforms, i.e.,

$$L_{\varepsilon, \Omega_v}(s_k) = E_{\varepsilon, \Omega_v}[\exp(-s_k I_G)] \quad (32)$$

$$L_{\Omega_A}(s_k) = E_{\Omega_A}[\exp(-s_k I_A)]. \quad (33)$$

According to [3], the Laplace transform of the small-scale fading of active interfering GBSs can further be expressed as (34), shown at the bottom of the next page, where (a) follows from $\varepsilon \sim B(p_\varepsilon)$ and the moment generating function (MGF) of a Bernoulli distribution; (b) follows from the MGF of a gamma distribution; (c) follows from:

$$\gamma(p_\varepsilon, m_v, s_k) = 1 - p_\varepsilon + p_\varepsilon \left[\frac{m_v}{m_v + s_k P_{GG}\xi_v(D_j)} \right]^{m_v}. \quad (35)$$

According to [2], the Laplace transform of the small-scale fading of all interfering ABSs can further be expressed as

$$\begin{aligned} L_{\Omega_A}(s_k) &= E_{\Omega_A} \left[\exp \left(- \sum_{i=1}^{N-1} s_k P_{AG} \Omega_A \xi_A(D_i) \right) \right] \\ &= \prod_{i=1}^{N-1} E_{\Omega_A} [\exp(-s_k P_{AG}^{\max} \cos(\varphi) \Omega_A \xi_A(D_i))] \end{aligned}$$

$$\begin{aligned} f_{D_A}(d_A|X_0) &= N(1 - F_{\tilde{D}_i}(d_A|X_0))^{N-1} f_{\tilde{D}_i}(d_A|X_0) \\ &= \begin{cases} N(1 - F_{\tilde{D}_{i,1}}(d_A|X_0))^{N-1} f_{\tilde{D}_{i,1}}(d_A|X_0), & 0 \leq d_A \leq R_A - X_0 \\ N(1 - F_{\tilde{D}_{i,2}}(d_A|X_0))^{N-1} f_{\tilde{D}_{i,2}}(d_A|X_0), & R_A - X_0 < d_A \leq R_A + X_0 \end{cases} \\ &= \begin{cases} N \left(\frac{2d_A}{R_A^2} \right) \left(\frac{R_A^2 - d_A^2}{R_A^2} \right)^{N-1}, & 0 \leq d_A \leq R_A - X_0 \\ N \left(\frac{2d_A}{\pi R_A^2} \theta^*(d_A) \right) \left[1 - \frac{d_A^2}{\pi R_A^2} \left(\theta^*(d_A) - \frac{1}{2} \sin(2\theta^*(d_A)) \right) - \frac{1}{\pi} \left(\phi^*(d_A) - \frac{1}{2} \sin(2\phi^*(d_A)) \right) \right]^{N-1}, & R_A - X_0 < d_A \leq R_A + X_0 \end{cases} \end{aligned} \quad (20)$$

$$\begin{aligned}
 &\stackrel{(a)}{=} \prod_{i=1}^{N-1} \left[\frac{m_L}{m_L + s_k P_A g_A^{\max} \left(\frac{\Delta h_{AU}}{\sqrt{\Delta h_{AU}^2 + D_i^2}} \right)^\omega \xi_L(D_i)} \right]^{m_L} \\
 &\stackrel{(b)}{=} \prod_{i=1}^{N-1} \psi(\omega, m_L, s_k)
 \end{aligned} \quad (36)$$

where (a) follows from the MGF of a gamma distribution; (b) follows from:

$$\psi(\omega, m_L, s_k) = \left[\frac{m_L}{m_L + s_k P_A g_A^{\max} \left(\frac{\Delta h_{AU}}{\sqrt{\Delta h_{AU}^2 + D_i^2}} \right)^\omega \xi_L(D_i)} \right]^{m_L}. \quad (37)$$

Thus, the conditional coverage probability of the typical AUE can be obtained by substituting (34) and (36) into (29), which is given by (38), shown at the bottom of the page.

D. Coverage Probability

The coverage probability and SINR meta distribution of a typical AUE depend on the b th moment of the conditional coverage probability of the AUE. According to [4], the b th moment of the conditional coverage probability is defined as

$$M_b = E_{\Phi_A, \Phi_G} \left[\left(P_{\text{cov}|X_0, \Phi_A, \Phi_G}(\delta) \right)^b \right], b \in N^* \quad (39)$$

where b denotes the order of the moment, N^* denotes a set of positive integers. M_b represents the expectation of the joint spatial distributions of the GBSs and ABSs in the network. Based on (38), M_b can further be expressed as (40), shown at the bottom of the next page, where $b_k, k = 1, 2, \dots, m_L$ is a nonnegative integer; (a) follows from the multinomial theorem [7]; (b) follows from the property of expectation; the term $F_1(b_k)$ and the term $F_2(b_k)$ can be expressed, respectively, as (41) and (42), shown at the bottom of the next page. In (41), (a) follows from the probability generating functional (PGFL) of an Non-HPPP [17]. In (42),

$$\begin{aligned}
 P_{\text{cov}|X_0, \Phi_A, \Phi_G}(\delta) &= P(\text{SINR} > \delta | X_0, \Phi_A, \Phi_G) \\
 &= P(P_{GG} \Omega_L \xi_L(D_G) + P_A g_A^{\max} \Omega_L \xi_L(D_A) > \delta(I_G + I_A + \sigma^2) | X_0, \Phi_A, \Phi_G) \\
 &\stackrel{(a)}{\approx} \sum_{k=1}^{m_L} \nu_k E_{\Omega_A, \Omega_v, \varepsilon} \left[\exp \left(- \frac{k \delta \tau}{P_{GG} \xi_L(D_G) + P_A g_A^{\max} \xi_L(D_A)} (I_G + I_A + \sigma^2) \right) \right] \\
 &= \sum_{k=1}^{m_L} \nu_k \exp(-s_k \sigma^2) \cdot E_{\varepsilon, \Omega_v} [\exp(-s_k I_G)] \cdot E_{\Omega_L} [\exp(-s_k I_A)] \\
 &\stackrel{(b)}{=} \sum_{k=1}^{m_L} \nu_k \exp(-s_k \sigma^2) L_{\varepsilon, \Omega_v}(s_k) L_{\Omega_L}(s_k).
 \end{aligned} \quad (29)$$

$$\begin{aligned}
 L_{\varepsilon, \Omega_v}(s_k) &= E_{\varepsilon, \Omega_v} \left[\exp \left(\sum_{\Phi_G \setminus \Phi'_B} -s_k \varepsilon P_{GG} \Omega_v \xi_v(D_j) \right) \right] \\
 &= E_{\varepsilon, \Omega_v} \left[\exp \left(\sum_{\Phi_G^L \setminus \Phi'_B} -s_k \varepsilon P_{GG} \Omega_L \xi_L(D_j) \right) \cdot \exp \left(\sum_{\Phi_G^{\bar{L}} \setminus \Phi'_B} -s_k \varepsilon P_{GG} \Omega_{\bar{L}} \xi_{\bar{L}}(D_j) \right) \right] \\
 &= \prod_{\Phi_G^L \setminus \Phi'_B} E_{\varepsilon, \Omega_L} [\exp(-s_k \varepsilon P_{GG} \Omega_L \xi_L(D_j))] \prod_{\Phi_G^{\bar{L}} \setminus \Phi'_B} E_{\varepsilon, \Omega_{\bar{L}}} [\exp(-s_k \varepsilon P_{GG} \Omega_{\bar{L}} \xi_{\bar{L}}(D_j))] \\
 &\stackrel{(a)}{=} \prod_{\Phi_G^L \setminus \Phi'_B} \{1 - p_\varepsilon + p_\varepsilon \cdot E_{\Omega_L} [\exp(-s_k P_{GG} \Omega_L \xi_L(D_j))]\} \prod_{\Phi_G^{\bar{L}} \setminus \Phi'_B} \{1 - p_\varepsilon + p_\varepsilon \cdot E_{\Omega_{\bar{L}}} [\exp(-s_k P_{GG} \Omega_{\bar{L}} \xi_{\bar{L}}(D_j))]\} \\
 &\stackrel{(b)}{=} \prod_{\Phi_G^L \setminus \Phi'_B} \left\{ 1 - p_\varepsilon + p_\varepsilon \left[\frac{m_L}{m_L + s_k P_{GG} \xi_L(D_j)} \right]^{m_L} \right\} \prod_{\Phi_G^{\bar{L}} \setminus \Phi'_B} \left\{ 1 - p_\varepsilon + p_\varepsilon \left[\frac{m_{\bar{L}}}{m_{\bar{L}} + s_k P_{GG} \xi_{\bar{L}}(D_j)} \right]^{m_{\bar{L}}} \right\} \\
 &\stackrel{(c)}{=} \prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k) \prod_{\Phi_G^{\bar{L}} \setminus \Phi'_B} \gamma(p_\varepsilon, m_{\bar{L}}, s_k)
 \end{aligned} \quad (34)$$

$$P_{\text{cov}|X_0, \Phi_A, \Phi_G}(\delta) = \sum_{k=1}^{m_L} \nu_k \exp(-s_k \sigma^2) \prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k) \prod_{\Phi_G^{\bar{L}} \setminus \Phi'_B} \gamma(p_\varepsilon, m_{\bar{L}}, s_k) \prod_{i=1}^{N-1} \psi(\omega, m_L, s_k). \quad (38)$$

(a) follows from that each point in a BPP is i.i.d. [17]; (b) is obtained by averaging the horizontal distances between the typical AUE and its interfering ABSs. Therefore, if $b = 1$, we can obtain the l th moment of the conditional coverage probability, i.e., the coverage probability, which is given by (43), shown at the bottom of the page.

E. SINR Meta Distribution

According to [26], the SINR meta distribution of a typical AUE is defined as the CCDF of the conditional coverage probability of the AUE. Let $F_{\text{meta}|X_0}(\delta, x)$ denote the SINR meta distribution. Thus, we have

$$F_{\text{meta}|X_0}(\delta, x) = P(P_{\text{cov}|X_0, \Phi_A, \Phi_G}(\delta) > x), \quad x \in [0, 1] \quad (44)$$

where x denotes the reliability threshold of an individual link. According to [4] and [5], for a network that follows an HPPP, the SINR meta distribution can be interpreted as the fraction of the links that achieve a

link reliability of at least x for an SINR threshold in each network realization. It can reflect more fine-grained information, such as link reliability about the individual links, compared with the coverage probability. However, it is difficult to derive the SINR meta distribution based on the Gil-Pelaez inversion theorem [27]. A standard beta distribution-based approach is usually used to approximate the SINR meta distribution by matching the first and second moments of the conditional coverage probability [4], [26]. Accordingly, the SINR meta distribution of a typical AUE is given by

$$F_{\text{meta}|X_0}(\delta, x) \approx 1 - I_x\left(\frac{M_1\beta}{1-M_1}, \beta\right) \quad (45)$$

where $\beta = [(M_1 - M_2)(1 - M_1)]/[M_2 - (M_1)^2]$; M_1 and M_2 denote the first and second moments, respectively, which are given by (40); and $I_x(a_I, b_I)$ is the CDF of a standard beta distribution with parameters a_I and b_I , which is given by

$$\begin{aligned} M_b &= E_{\Phi_A, \Phi_G} \left[(P_{\text{cov}|X_0, \Phi_A, \Phi_G}(\delta))^b \right] \\ &= E_{\Phi_A, \Phi_G} \left[\left(\sum_{k=1}^{m_L} v_k \exp(-s_k \sigma^2) \prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k) \prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k) \prod_{i=1}^{N-1} \psi(\omega, m_L, s_k) \right)^b \right] \\ &\stackrel{(a)}{=} \sum_{b_1 + \dots + b_{m_L} = b} \frac{b! E_{\Phi_A, \Phi_G} \left[\prod_{k=1}^{m_L} v_k^{b_k} \exp(-b_k s_k \sigma^2) \prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k)^{b_k} \prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k)^{b_k} \prod_{i=1}^{N-1} \psi(\omega, m_L, s_k)^{b_k} \right]}{b_1! \dots b_{m_L}!} \\ &\stackrel{(b)}{=} \sum_{b_1 + \dots + b_{m_L} = b} \frac{b! E_{D_A, D_G} \left[\prod_{k=1}^{m_L} v_k^{b_k} \exp(-b_k s_k \sigma^2) E_{D_j} \left[\prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k)^{b_k} \prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k)^{b_k} \right] E_{D_i} \left[\prod_{i=1}^{N-1} \psi(\omega, m_L, s_k)^{b_k} \right] \right]}{b_1! \dots b_{m_L}!} \\ &= \sum_{b_1 + \dots + b_{m_L} = b} \frac{b! E_{D_A, D_G} \left[\prod_{k=1}^{m_L} v_k^{b_k} \exp(-b_k s_k \sigma^2) \cdot F_1(b_k) \cdot F_2(b_k) \right]}{b_1! \dots b_{m_L}!} \\ &= \sum_{b_1 + \dots + b_{m_L} = b} \frac{b!}{b_1! \dots b_{m_L}!} \left\{ \int_0^\infty \left\{ \int_0^{R_A - X_0} \left[\prod_{k=1}^{m_L} v_k^{b_k} \exp(-b_k s_k \sigma^2) \cdot F_1(b_k) \cdot F_2(b_k) \right] \cdot f_{D_A, D_G}(d_A, d_G | X_0) dd_A \right. \right. \\ &\quad \left. \left. + \int_{R_A - X_0}^{R_A + X_0} \left[\prod_{k=1}^{m_L} v_k^{b_k} \exp(-b_k s_k \sigma^2) \cdot F_1(b_k) \cdot F_2(b_k) \right] \cdot f_{D_A, D_G}(d_A, d_G | X_0) dd_A \right\} dd_G \right\} \quad (40) \end{aligned}$$

$$\begin{aligned} F_1(b_k) &= E_{R_j} \left[\prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k)^{b_k} \prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k)^{b_k} \right] = E_{R_j} \left[\prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k)^{b_k} \right] \cdot E_{R_j} \left[\prod_{\Phi_G^L \setminus \Phi'_B} \gamma(p_\varepsilon, m_L, s_k)^{b_k} \right] \\ &\stackrel{(a)}{=} \exp \left(- \int_{d_G + w}^\infty (1 - \gamma(p_\varepsilon, m_L, s_k)^{b_k}) \cdot 2\pi \lambda_G^L d_j dd_j \right) \exp \left(- \int_{d_G + w}^\infty (1 - \gamma(p_\varepsilon, m_L, s_k)^{b_k}) \cdot 2\pi \lambda_G^L d_j dd_j \right) \\ &= \exp \left(-2\pi \lambda_G \int_{d_G + w}^\infty d_j (1 - P_{\text{LoS}} \gamma(p_\varepsilon, m_L, s_k)^{b_k} - P_{\text{NLoS}} \gamma(p_\varepsilon, m_L, s_k)^{b_k}) dd_j \right). \quad (41) \end{aligned}$$

$$\begin{aligned} F_2(b_k) &= E_{D_i} \left[\prod_{i=1}^{N-1} \psi(\omega, m_L, s_k)^{b_k} \right] \stackrel{(a)}{=} \left\{ E_{D_i} \left[\psi(\omega, m_L, s_k)^{b_k} \right] \right\}^{N-1} \stackrel{(b)}{=} \left\{ \int_{D_i} \psi(\omega, m_L, s_k)^{b_k} \cdot f_{D_i}(d_i | d_A, x_0) dd_i \right\}^{N-1} \\ &= \begin{cases} \left[\int_{d_A}^{R_A - X_0} \psi(\omega, m_L, s_k)^{b_k} \cdot \frac{f_{R_{i,1}}(d_i | X_0)}{1 - F_{R_{i,1}}(d_A | X_0)} dd_i + \int_{R_A - X_0}^{R_A + X_0} \psi(\omega, m_L, s_k)^{b_k} \cdot \frac{f_{R_{i,2}}(d_i | X_0)}{1 - F_{R_{i,1}}(d_A | X_0)} dd_i \right]^{N-1}, & 0 \leq d_A \leq R_A - X_0 \\ \left[\int_{R_A - X_0}^{R_A + X_0} \psi(\omega, m_L, s_k)^{b_k} \cdot \frac{f_{R_{i,2}}(d_i | X_0)}{1 - F_{R_{i,1}}(d_A | X_0)} dd_i \right]^{N-1}, & R_A - X_0 < d_A \leq R_A + X_0 \end{cases} \quad (42) \end{aligned}$$

$$\begin{aligned} P_{\text{cov}|X_0}(\delta) &= M_1 = P(\text{SINR} > \delta | X_0) = \sum_{k=1}^{m_L} \left\{ \int_0^\infty \left\{ \int_0^{R_A - X_0} v_k \exp(-s_k \sigma^2) \cdot F_1(1) \cdot F_2(1) \cdot f_{D_A, D_G}(d_A, d_G | X_0) dd_A \right. \right. \\ &\quad \left. \left. + \int_{R_A - X_0}^{R_A + X_0} v_k \exp(-s_k \sigma^2) \cdot F_1(1) \cdot F_2(1) \cdot f_{D_A, D_G}(d_A, d_G | X_0) dd_A \right\} dd_G \right\} \quad (43) \end{aligned}$$

$$I_x(a_I, b_I) = \frac{\int_0^x s^{a_I-1} (1-s)^{b_I-1} ds}{B(a_I, b_I)}, \quad a_I, b_I > 0 \quad (46)$$

$$\text{with } B(a_I, b_I) = \int_0^1 s^{a_I-1} (1-s)^{b_I-1} ds. \quad (47)$$

F. Achievable Throughput

The achievable throughput of a typical AUE is defined as the maximum bit rate that an AUE can obtain in the network [18]. According to [3], the achievable throughput of a typical AUE is given by

$$\begin{aligned} C &= E[\log_2(1 + \text{SINR})] \\ &= \frac{1}{\ln 2} \int_0^\infty P(\text{SINR} > e^x - 1) dx \\ &= \frac{1}{\ln 2} \int_0^\infty \frac{P_{\text{cov}|X_0}(\delta)}{\delta + 1} d\delta. \end{aligned} \quad (48)$$

However, (48) contains an integration operation. According to [18], we can use the Gauss–Chebyshev Quadrature to approximately obtain the achievable throughput, i.e.,

$$C = \frac{1}{\ln 2} \sum_{k=1}^K \frac{P_{\text{cov}|X_0}(\chi_k)}{1 + \chi_k} \cdot \frac{\pi^2 \sin\left(\frac{2k-1}{2K}\pi\right)}{4K \cdot \cos^2\left[\frac{\pi}{4} \cos\left(\frac{2k-1}{2K}\pi\right) + \frac{\pi}{4}\right]} \quad (49)$$

where $\chi_k = \tan[(\pi/4) \cos((2k-1)/2K)\pi] + (\pi/4)$, and K is an adjustable integer and is supposed to be large enough to ensure the accuracy of this approximation.

VI. PERFORMANCE EVALUATION

This section first validates the derived performance models through simulation results, and then evaluates the performance of the proposed AG_CoMP strategy through numerical results. In the evaluation, we introduce a 2-GBS cooperation strategy, a 2-ABS cooperation strategy and a simple aerial and ground BS cooperation (AG_Co) strategy as benchmarks for comparison. For the 2-GBS cooperation strategy, no ABS participates in BS cooperation, only the closest GBS and the second closest GBS are associated to provide downlink transmission for an AUE through noncoherent joint transmission. For the 2-ABS cooperation strategy, no GBS participates in BS cooperation, only the closest ABS and the second closest ABS are associated to provide downlink transmission for an AUE through noncoherent joint transmission. The setting of the blanking area in the 2-GBS cooperation strategy and the 2-ABS cooperation strategy is the same as the proposed AG_CoMP strategy. For the AG_Co strategy, the closest GBS and the closest ABS to the AUE are associated to provide downlink transmission for an AUE through noncoherent joint transmission, and there is no blanking area. Monte Carlo simulation experiments were conducted on a MATLAB platform. According to [3] and [4], we consider a circular area with radius R_{sim} large enough as the GBS distribution area in the network. The values of the main parameters used to obtain the numerical results are given in Table I unless otherwise stated.

TABLE I
MAIN PARAMETERS

Parameter	Value
λ_G	20 km ⁻² [3]
(h_G, h_A)	(25, 500) m
N	20
(P_G, P_A)	(43, 30) dBm [2]
$(g_G, g_A^{\max})$	(-15, 0) dB [2]
(ρ_L, ρ_L^-)	(-41.1, -32.9) dB [3][28]
$(\alpha_L, \alpha_L^-, m_L, m_L^-)$	(2.09, 3.75, 3, 1) [3][28]
$(a_{LoS}, b_{LoS}, c_{LoS})$	(1, 0.106, 1) [2]
(ω, p_ε)	(2, 0.9)
δ	-8 dB
(X_0, w)	(200, 50) m
σ^2	-113 dBm [2]
(R_A, R_{sim})	(1, 3) km

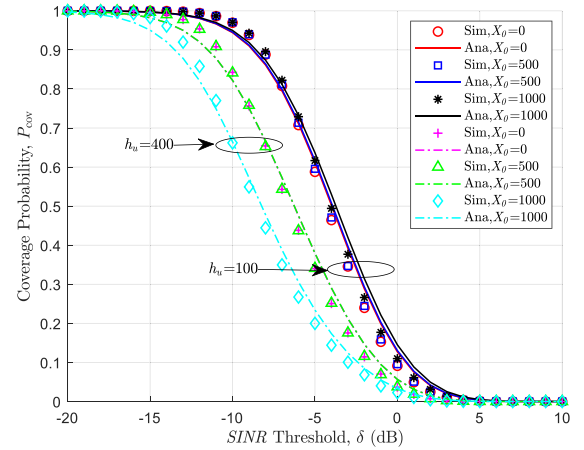


Fig. 3. Coverage probability versus SINR threshold for different X_0 and h_u .

Figs. 3–12 show the analytical results (Ana) and the simulation results (Sim). It is observed that they are very close to each other, which justifies the effectiveness of the derived performance models.

A. Coverage Probability

Fig. 3 shows the coverage probability of an AUE with the AG_CoMP strategy versus the SINR threshold for different values of X_0 and different AUE altitudes. It is seen that as the SINR threshold increases, the coverage probability gradually decreases. This is because as the SINR threshold increases, the SINR requirement for downlink transmission of the AUE increases, which makes a downlink transmission more likely to fail and thus reduces the coverage probability. Moreover, for $h_u = 100$ m, there is little difference in the coverage probability of an AUE for different values of X_0 ; for $h_u = 400$ m, the coverage probabilities for $X_0 = 0$ and $X_0 = 500$ have little difference. Only if the AUE is located at the edge of the ABS distribution area (i.e., $X_0 = 1000$), the

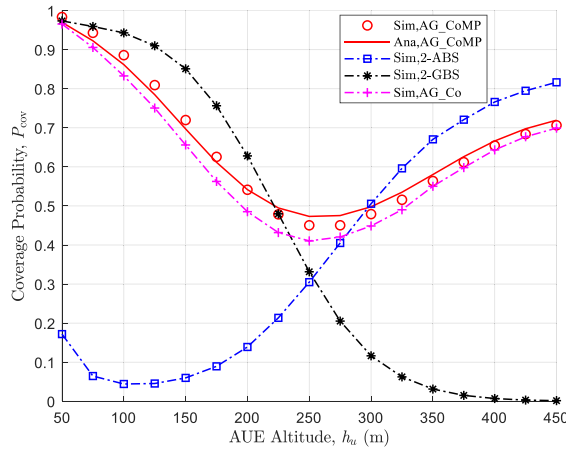


Fig. 4. Coverage probability versus AUE altitude.

coverage probability would be largely reduced. This means that if the AUE is close to an ABS (e.g., $h_u = 400$ m), the spatial location of the AUE has little impact on the coverage probability unless the AUE is located at the edge of the ABS distribution area. If the AUE is close to a GBS (e.g., $h_u = 100$ m), no matter whether the AUE is located at the edge of the ABS distribution area or not, there is little difference in the coverage probability of the AUE. This is because if the AUE is close to the GBS, compared with the ABS, the GBS has a major effect on the SINR of the AUE, and thus the value of X_0 has little impact on the coverage probability.

Fig. 4 shows the coverage probability of an AUE with the AG_CoMP strategy and those with the benchmark strategies for different AUE altitudes. It is seen that as the AUE altitude increases, the coverage probability with the AG_CoMP strategy first gradually decreases and then gradually increases, and it has a minimum value at a certain AUE altitude. This is because as the AUE altitude increases, the distance between the AUE and the GBS gradually increases, while the distance between the AUE and the ABS gradually decreases. When the AUE altitude is smaller, the AUE is closer to a GBS than an ABS. In this case, the useful signal power from the cooperative GBS is large enough to dominate. As the AUE altitude increases, the useful signal power from the cooperative GBS gradually decreases, making the SINR received at the AUE decrease and the coverage probability decrease until the useful signal power from the cooperative GBS is small enough to no longer dominate. When the AUE altitude is larger, the AUE becomes closer to an ABS than a GBS. In this case, the useful signal power from the cooperative ABS gradually increases and exceeds that from the cooperative GBS, making it dominate in the SINR received at the AUE, and thus resulting in an increase in the SINR received at the AUE and accordingly in the coverage probability. On the other hand, when the AUE altitude is smaller, the coverage probability with the AG_CoMP strategy is a little smaller than that with the 2-GBS strategy, much larger than that with the 2-ABS strategy. When the AUE altitude is larger, the coverage probability with the AG_CoMP strategy is a little smaller than that with the 2-ABS strategy, much larger than that with the 2-GBS strategy. Especially, when the AUE is

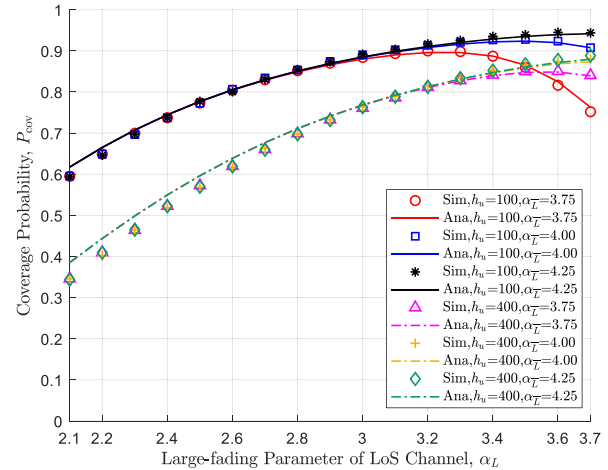


Fig. 5. Coverage probability versus large-fading parameter of a LoS channel.

at an intermediate altitude between an ABS and a GBS, the coverage probability with the AG_CoMP strategy is larger than those with the 2-GBS strategy and the 2-ABS strategy. For example, when $h_u = 250$ m, the coverage probability with the AG_CoMP strategy can be, respectively, increased by 42.75% and 55.20% compared with those with the 2-GBS strategy and the 2-ABS strategy. These indicate that the AG_CoMP strategy can exploit the advantages of the ABS and GBS cooperation to ensure a relatively large coverage probability of the AUE regardless of the AUE's altitude, and can effectively improve the coverage probability of the AUE when the AUE is at an intermediate altitude between an ABS and a GBS. Moreover, the coverage probability with the AG_CoMP strategy is larger than that with the AG_Co strategy regardless of UAV altitudes. This indicates that the blanking area can further improve the coverage probability.

Fig. 5 shows the coverage probability of an AUE with the AG_CoMP strategy versus the large-fading parameter of a LoS channel (α_L) for different large-fading parameters of an NLoS channel (α_T) and different AUE altitudes. It is seen that as α_L increases, the coverage probability of an AUE first gradually increases and then gradually decreases. This is because as α_L increases, the path loss of the LoS channel gradually decreases. When the value of α_L is smaller, the decrease in the LoS interference received by an AUE dominates relative to the decrease in the useful signal power received by the AUE, thus resulting in an increase in the SINR received at the AUE and accordingly in the coverage probability. When the value of α_L increases to be close to the value of α_T , the decrease in the useful signal power received by an AUE dominates relative to the decrease in the LoS interference received by an AUE, thus resulting in a decrease in the SINR received at the AUE and accordingly in the coverage probability. On the other hand, when the value of α_L is much smaller than the value of α_T , the coverage probability of an AUE almost does not change for different values of α_T . When the value of α_L is close to the value of α_T , a larger value of α_T results in a larger value of the coverage probability of an AUE. This indicates that the NLoS interference received by an AUE cannot be ignored when the value of α_L is close to the value of α_T .

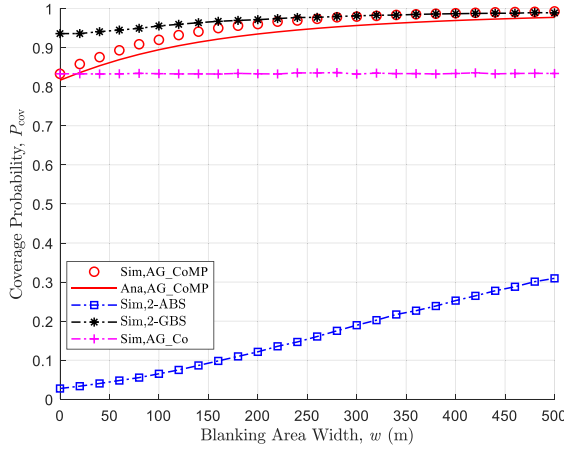
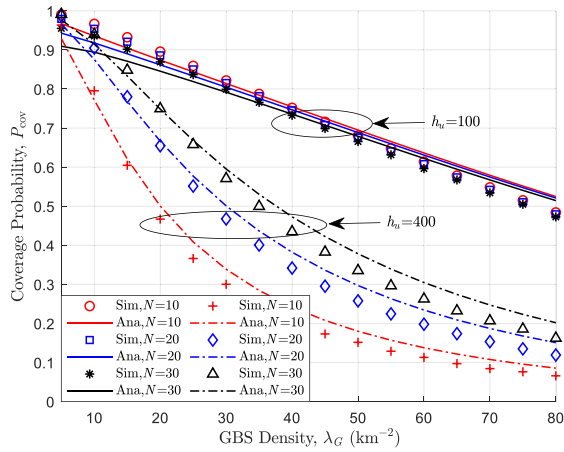

 Fig. 6. Coverage probability versus blanking area width ($h_u = 100$ m).

 Fig. 7. Coverage probability versus GBS density for different h_u and N .

Fig. 6 shows the coverage probability of an AUE with the AG_CoMP strategy and those with the benchmark strategies for different values of the blanking area width. It is seen that as the blanking area width increases, the coverage probabilities with the AG_CoMP strategy, the 2-GBS strategy and the 2-ABS strategy gradually increase, and the coverage probability with the AG_Co strategy remains the same. This is because in this case, for the AG_CoMP strategy, the 2-GBS strategy and the 2-ABS strategy, the number of the GBSs that use the same time-spectrum resource blocks used by the AUE decreases, which increases the SINR and coverage probability; while the AG_Co strategy does not set a blanking area, which makes the coverage probability unchanged with the increase of the blanking area width. This also indicates that a moderate blanking area width can effectively improve the coverage probability. On the other hand, when the blanking area width becomes larger, the coverage probabilities with the AG_CoMP strategy and the 2-GBS strategy are close and tend to be closer, and are larger than those with the 2-ABS strategy and the AG_Co strategy. This means that when the AUE altitude is smaller, we can appropriately increase the blanking area width with the AG_CoMP strategy to make the coverage probability to reach that with the 2-GBS strategy.

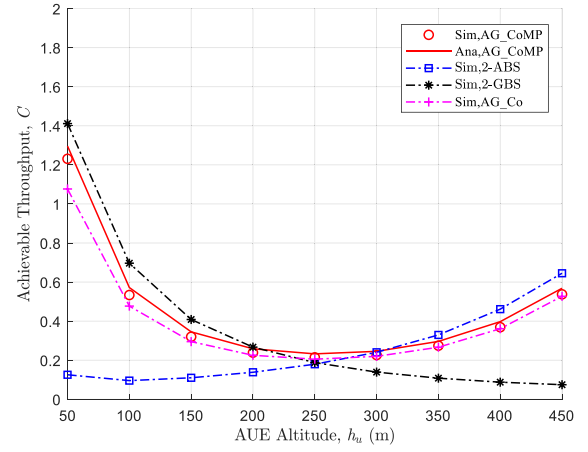


Fig. 8. Achievable throughput versus AUE altitude.

Fig. 7 shows the coverage probability of an AUE with the AG_CoMP strategy versus the GBS density for different AUE altitudes and different numbers of ABSs. It is seen that as the GBS density increases, the coverage probability gradually decreases. This is because as the GBS density increases, the number of interfering GBSs increases and the distance between the AUE and the cooperative GBS decreases. Compared with the increase of the useful signal power caused by the decreased distance, the increase of the interference signal power caused by the increased number of interfering GBSs will dominate, which reduces the coverage probability. On the other hand, for $h_u = 100$ m, a larger number of ABSs results in a smaller value of the coverage probability; for $h_u = 400$ m, a larger number of ABSs results in a larger value of the coverage probability. This is because if the AUE is close to the GBS (e.g., $h_u = 100$ m), as the number of ABSs increases, compared with the increase of the useful signal power caused by the decreased horizontal distance between the AUE and the cooperative ABS, the increase of the interference signal power caused by the increased number of interfering ABSs dominates, which reduces the coverage probability as well. On the contrary, if the AUE is close to the ABS (e.g., $h_u = 400$ m), as the number of ABSs increases, compared with the increase of the interference signal power caused by the increased number of interfering ABSs, the increase of the useful signal power caused by the decreased horizontal distance between the AUE and the cooperative ABS dominates, which increases the coverage probability.

B. Achievable Throughput

Fig. 8 shows the achievable throughput of an AUE with the AG_CoMP strategy and those with the benchmark strategies for different AUE altitudes. It is seen that similar to the coverage probability of an AUE shown in Fig. 4, the AG_CoMP strategy can ensure a relatively large achievable throughput of the AUE regardless of the AUE's altitude, and can effectively improve the achievable throughput of the AUE at an intermediate altitude between an ABS and a GBS, and a blanking area can further improve the achievable throughput.

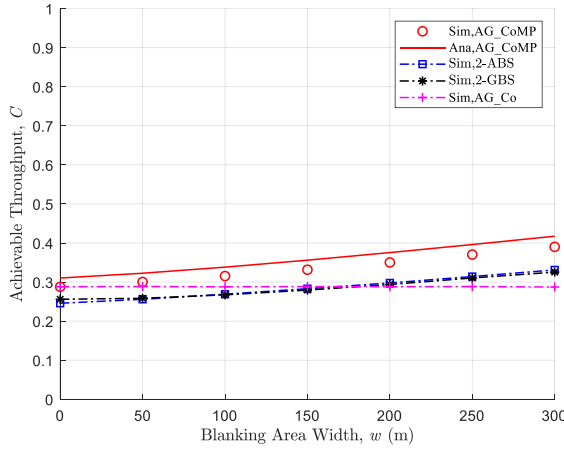


Fig. 9. Achievable throughput versus blanking area width ($h_u = 250$ m, $p_e = 0.6$).

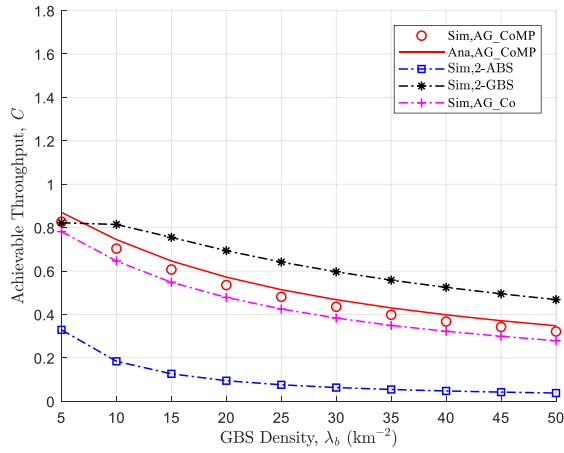


Fig. 10. Achievable throughput versus GBS density ($h_u = 100$ m).

Fig. 9 shows the achievable throughput of an AUE with the AG_CoMP strategy and those with the benchmark strategies for different values of the blanking area width. It is seen that as the blanking area width increases, the achievable throughputs with the AG_CoMP strategy, the 2-GBS strategy and the 2-ABS strategy gradually increase, and the achievable throughput with the AG_Co strategy remains the same. This reason is similar to that for the coverage probability of the AUE explained in Fig. 6. This indicates that the achievable throughput of the AUE can be efficiently improved by setting a moderate blanking area width. On the other hand, when $h_u = 250$ m, the achievable throughput with the AG_CoMP strategy is larger than those with the benchmark strategies. This also indicates the superiority of the proposed AG_CoMP strategy for an AUE at an intermediate altitude.

Fig. 10 shows the achievable throughput of an AUE with the AG_CoMP strategy and those with the benchmark strategies for different GBS densities. It is seen that for $h_u = 100$ m, when the GBS density is smaller (e.g., $\lambda_g = 5$ km⁻²), the achievable throughput with the AG_CoMP strategy is larger than that with the 2-GBS strategy. This indicates that when the GBS density is smaller, even if the AUE altitude

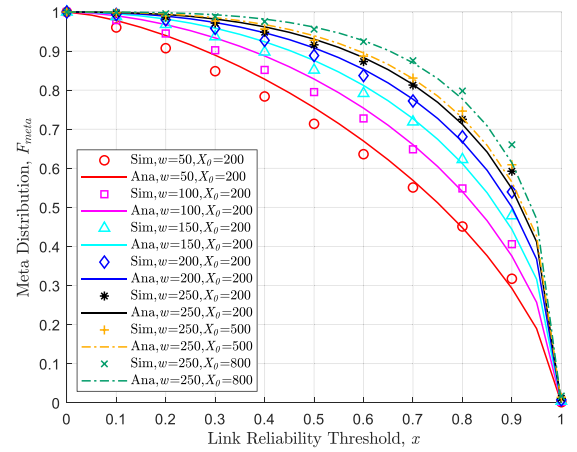


Fig. 11. Meta distribution versus link reliability threshold for different w and X_0 . ($\delta = -2$ dB, $h_u = 50$ m).

is low, compared with the 2-GBS strategy, the AUE with the AG_CoMP strategy can still obtain a larger achievable throughput.

C. SINR Meta Distribution

Fig. 11 shows the SINR meta distribution of an AUE with the AG_CoMP strategy versus the link reliability threshold for different values of the blanking area width and X_0 . It is seen that as the link reliability threshold increases, the SINR meta distribution of an AUE gradually decreases. This is because as the link reliability threshold increases, the requirement for the downlink reliability of an AUE increases, which decreases the SINR meta distribution. On the other hand, a larger value of the blanking area width results in a larger value of the SINR meta distribution. This reason is similar to that for the coverage probability of an AUE explained in Fig. 6. This indicates that an appropriate value of the blanking area width can efficiently improve the SINR meta distribution of an AUE. Moreover, a larger value of X_0 results in a larger value of the SINR meta distribution, which is different from the coverage probability in Fig. 3. This is because as the value of X_0 increases, the AUE become closer to the edge of the ABS distribution area, which reduces the interference from ABSs and thus increases the SINR meta distribution. This indicates that compared to the coverage probability, the SINR meta distribution can reflect the impact of the nondominant interference from ABSs on the SINR, and demonstrates that the SINR meta distribution can provide more fine-grained information on the SINR.

Fig. 12 shows the SINR meta distribution of an AUE with the AG_CoMP strategy, the 2-GBS strategy and the 2-ABS strategy versus the SINR threshold for different AUE altitudes. It is seen that as the SINR threshold increases, the SINR meta distributions with all the strategies gradually decreases. This reason is similar to that for the coverage probability of an AUE explained in Fig. 3. On the other hand, for $h_u = 400$ m, the SINR meta distribution with the AG_CoMP strategy is much larger than that with the 2-GBS strategy, and smaller than that with the 2-ABS strategy; for $h_u = 100$ m, the SINR meta

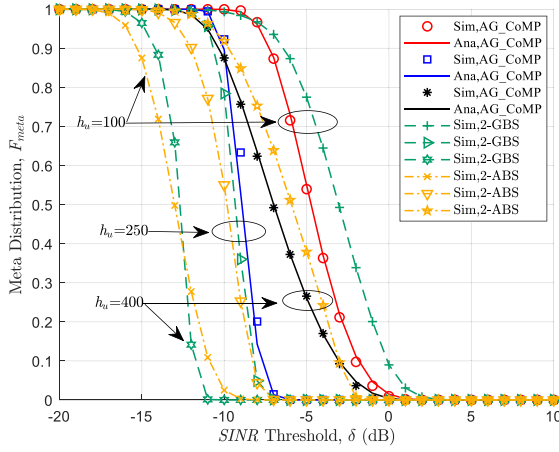


Fig. 12. Meta distribution versus SINR threshold ($x = 0.6$).

distribution with the AG_CoMP strategy is much larger than that with the 2-ABS strategy, and smaller than that with the 2-GBS strategy; for $h_u = 250$ m, the SINR meta distribution with the AG_CoMP strategy is larger than those with the 2-GBS strategy and the 2-ABS strategy. This is similar to those for the coverage probability and the achievable throughput of an AUE in Fig 4 and Fig. 8, and thus indicates that the proposed AG_CoMP strategy can efficiently improve the SINR meta distribution of the AUE, especially at an intermediate altitude between an ABS and a GBS.

VII. CONCLUSION

In this article, we studied BS cooperation in a Ucell_Net and proposed an AG_CoMP strategy for improving the downlink transmission performance of an AUE in the network. The proposed AG_CoMP strategy allows both a GBS and an ABS to participate in BS cooperation, and seeks to reduce the negative impact on the overall network performance. The GBS and ABS closest to an AUE are associated to cooperatively provide downlink transmission of the AUE and a blanking area is defined to disallow nearby GBSs to transmit during the transmission of the AUE. A combination of noncoherent joint transmission and DPB is used to reduce the large signaling overhead for transmission of the AUE's data between the cooperative BSs with the proposed strategy. Performance models were derived to analyze the downlink transmission performance of an AUE in terms of the coverage probability, the achievable throughput, and the SINR meta distribution of the AUE. The numerical results showed that compared with the three benchmark strategies, the proposed AG_CoMP strategy can ensure a relatively good downlink transmission performance, and can achieve a better downlink transmission performance when the AUE is at an intermediate altitude between an ABS and a GBS. For future work, we intend to investigate the impacts of the blanking area width, the AUE mobility, and a nonperfect wireless backhaul link on the downlink transmission performance. Moreover, we will also study the optimization issue of the number of cooperative ABSs and GBSs with the proposed AG_CoMP strategy.

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